DATASCI 530 – Project 1 and 2

For this assignment you will be asked to create and analyze a dataset via web scraping. Pick an online marketplace where you can search for products. Your goal is to design a web scraping algorithm that does the following:

Project 1 Deliverables:

- Search for a website that you want to scrape.
- Conduct an automated search for a particular type of product. You choose which type of product you want to analyze.
- Extract information about products (name, vendor, price, rating, detailed product description, and product categories). You do not need to extract all these characteristics depending on your website. The data you extract should be stored in one or more datasets.
- Document information about your web scraping process: when the data was collected, how many observations you chose,
- Your algorithm should include navigation: There should be at least one instance of clicking on a button, going to another page, and extracting information from that page. Navigation should be automated via loops or parallelization functions.

Project 2 Deliverables:

- Address any comments to your project 1 deliverable.
- With the data collected, produce key descriptive statistics.
- In your report include an analysis of what we learn from this data.

You will have to submit an HTML file with your results and a "csv" file with the data that you extracted.

Rubric for Project 1:

You will be graded both on the code and the writing using the following point system.

Component	Detailed Points	Total Points
Project Description Overview: What is the source of the data, when the data was collected, what data within the website was collected, and decisions about how to restrict the data collection process.	3	3
Scraping Algorithm		12
(a) Quality of code (no errors, proper use of web scraping tools)	2	
(b) Automate data collection via loops	3	
(c) Extract data from multiple items (for example, in a marketplace it would be different products)	2	
(d) Algorithms includes navigation (there is an instance on clicking of a button or link, going to another page, and extracting information)	2	
(e) Covert raw data to a workable dataset (dataset is included with submission).	3	
Results		4
(a) Quality Checks (Count number of observations, check that data is non-empty)	2	
(b) Present a table counting the number of observations and missing values. Present mean and variance for any numeric variables.	2	
Discussion Clarity and conciseness	1	1
Total		20

Rubric for Project 2:

You will be graded both on the code and the writing using the following point system. The second deliverable should primarily focus on (i) addressing any feedback from Project 1, (ii) producing a report.

The report should import the dataset from Project 1 in a separate Jupyter notebook and present final results intended for a non-coder audience.

Component	Detailed Points	Total Points
Overall		3
(a) Organizing and documenting code	2	
(c) Aesthetics	1	
Introduction		2
(a) Project Description	1	
(b) Summarize findings (There should be 3 or 4 key findings summarized at the beginning of document)	1	
Data Collection Process		4
Address feedback from Project 1 deliverable	4	
Results:		10
(a) Table to present information. This should include tables of summary statistics, number of observations.	4	
(b) Figures to present information, which include description of any categorical or numeric data. There should be a high focus on interpreting insights from the data.	6	
Discussion		
Clarity and conciseness	1	1
Total		20